Bayesian Epistemic Uncertainty for Reasoning Trajectory Selection in Language Model Fine-Tuning

Introduction and Motivation

Large language models demonstrate significantly improved performance in mathematical and multi-step reasoning when trained to generate intermediate reasoning steps rather than directly predicting the final answer (Wei et al., 2022). This establishes the foundation of many works that use reasoning trajectory as training data, including bootstrap rationales for hard problems (Zelikman et al., 2022) and methods that use a verifier to select the one ranked highest among many generated candidate solutions. In this method, the quality of selected reasoning trajectories is crucial. If a stronger teacher model can generate several different step-by-step reasoning solutions to each math problem, a student model may learn more math knowledge and concepts compared to others. Utilizing the teacher model's reasoning dataset is essential due to benchmarks such as GSM8K and MATH manifests that mathematical reasoning remains challenging even for large modern transformer models (Cobbe et al., 2021; Hendrycks et al., 2021).

The core idea is to prefer training data that the student model finds uncertain, because those samples may contain information that the student model has not yet learned. Nonetheless, uncertainty is the clue that we can use to identify which datasets have the most information that the model lacks. Kendall and Gal (2017) distinguish uncertainty into two types. Aleatoric uncertainty, which comes from ambiguity or variability in data, and epistemic uncertainty, which reflects a lack of knowledge in the model itself. For reasoning trajectories, this difference is significant. A student model may be assigned high uncertainty (entropy) from the generated solution by the teacher model since the underlying mathematical concept is unfamiliar, but it may also do the same thing due to the text generation using unusual wording, formatting, or an uncommon reasoning path. If our data trajectory is selected based on total predicted entropy, then the data selection method may overemphasize noisy or semantic variable examples rather than the examples that focus on the math knowledge that better improve the student model's reasoning steps.

This project studies whether a modern Bayesian theory of uncertainty can improve reasoning-data selection. In the experiments, I compare two trajectory-selection methods for teacher-generated math solutions. The baseline rule selects the trajectory with the highest total predictive entropy, while the proposal method uses the rule that selects the trajectory with the highest epistemic uncertainty. To calculate the epistemic uncertainty, we implement the technique of Bayesian approximation by mutual information under Monte Carlo dropout (Gal and Ghahramani, 2016). In the experiment, a large teacher model generates multiple candidate trajectories for each math question. A Qwen2.5-3B student model then scores these trajectories and is subsequently fine-tuned via LoRA on the datasets produced by the two selection rules. The trained student models are then evaluated on GSM8K and MATH-500. The study further analyzes whether separating epistemic uncertainty from total uncertainty provides a more informative fine-tune dataset and leads to better mathematical reasoning performance.

Method

In this project, I use a teacher-student pipeline to compare two uncertainty-based rules for

reasoning trajectory selection. First, Qwen2.5-72B-Instruct serves as the teacher model and generates five candidate step-by-step solutions for each math training problem. The multiply generated trajectories provide distinct possible reasoning for the same question and let the student model determine which sample may be most informative for learning.

To score these trajectories, Qwen2.5-3B serves as the student model. Instead of treating it as a traditional fully deterministic model, I approximate Bayesian uncertainty using Monte Carlo dropout. During inference time, dropout remains active for scoring, but gradients are disabled to avoid weight updates.

For each candidate trajectory, the model performs 10 stochastic forward passes under teacher forcing, meaning that the model evaluates the fixed reasoning trace token by token. The repeated passes generate slightly different token probability distributions, which are used for estimating data information uncertainty for the student model.

From these outputs, I compute two token-level uncertainty measures. The first is total predictive entropy, which captures the overall uncertainty of the average predictive distribution. The second is mutual information (MI), computed as total entropy minus expected entropy across the 10 stochastic passes. In this project, total entropy is regarded as the baseline uncertainty, while MI is used as an approximation of epistemic uncertainty, since it represents disagreement across stochastic passes and better captures the knowledge the student model lacks rather than surface-level variability.

In order to cover the token-level uncertainty into a single trajectory score, I compute and rank the uncertainty values for each trajectory and average the top 50 tokens. This top-50 aggregation focuses on the most informative parts of the reasoning chain rather than diluting the score across many low-information tokens within a single generated solution. Each candidate trajectory receives both a total entropy score and an MI score.

These scores allow building separate fine-tuning datasets. In the baseline method, it selects the trajectory with the highest total entropy for each question. In the proposal method, it selects the trajectory with the highest MI. This produces two training datasets with different selection rules. These training sets are then used to fine-tune separate LoRA adapters on the Qwen2.5-3B base model and evaluate the trained models on GSM8K and MATH-500. This experiment and setup control the effect of the trajectory selection and compare the total predictive entropy and epistemic uncertainty for reasoning-data curation.

Experimental Setup

The training pipeline began with 12,000 unique math problems. For each problem, the teacher model, Qwen2.5-72B-Instruct, generated five candidate reasoning trajectories, producing 60,000 total candidate solutions. Each candidate was then scored by a Qwen2.5-3B student model with Monte Carlo dropout implementation. For each math problem, I selected one trajectory with the highest total predictive entropy to consist of the baseline dataset and one trajectory with the highest mutual information (MI) to consist of the epistemic dataset. Each of the two final fine-tune datasets contains 12,000 samples with only a difference in the trajectory-selection rule.

To compare the total predictive entropy and MI of datasets, I fine-tuned two LoRA adapters on the same base student model, Qwen2.5-3B, using identical hyperparameters. The LoRA configuration used rank 16, LoRA alpha 32, and dropout. Training was conducted using 2 epochs with a per-device batch size of 4, gradient accumulation of 4, learning rate 2×10^{-4}, and maximum sequence length of 4096.

After the training procedure, both fine-tune models were evaluated on two standard mathematical reasoning benchmarks: MATH-500 and GSM8K. During the evaluation process, the models generated a single response for each math question. I then evaluate the models by extracting the first boxed answer from the generation and mark the prediction for the question correctly if the extracted string exactly matches the ground truth. The ACC@1 metric was used to compute and represent the model's accuracy on both benchmarks. The reference answers for GSM8K and MATH-500 were numeric solutions and open-ended answers, respectively. One limitation of the benchmark method is that using exact string matching will penalize formatting differences even when the underlying answer is mathematically correct.

Results

Table 1 shows the experiment results. The model trained on the MI dataset slightly outperformed the model trained on the total-entropy dataset in both benchmarks. On MATH-500, accuracy increased from 46.20% to 46.40%. On GSM8K, the accuracy increased from 79.30% to 79.45%. Although these gains are small, they are consistent across both evaluation results.

Table 1. Overall benchmark accuracy (strict ACC@1)

Benchmark	**Baseline (Total Entropy)**	**Epistemic (MI)**
MATH-500	46.20% (231/500)	46.40% (232/500)
GSM8K	79.30% (1046/1319)	79.45% (1048/1319)

The MATH-500 subject breakdown (Table 2) shows that the MI model helped in some areas but not all. It improves Counting and Probability from 34.2% to 36.8%, Intermediate Algebra from 25.8% to 30.9%, Number Theory from 71.0% to 74.2%, and Precalculus from 16.1% to 19.6%. However, it was slightly worse on Algebra, dropping from 65.3% to 64.5%, and more noticeably worse on Geometry, where accuracy fell from 36.6% to 29.3%, and on Prealgebra, where accuracy fell from 53.7% to 47.6%. This pattern suggests that the MI selection rule did not uniformly improve all types of reasoning, but it may have been useful in situations where the student model had larger knowledge gaps.

Table 2. MATH-500 accuracy by subject

Subject	**Total**	**Baseline Accuracy**	**Epistemic (MI) Accuracy**
Algebra	124	65.3% (81/124)	64.5% (80/124)
Counting & Probability	38	34.2% (13/38)	36.8% (14/38)
Geometry	41	36.6% (15/41)	29.3% (12/41)
Intermediate Algebra	97	25.8% (25/97)	30.9% (30/97)

Subject	Total	Baseline Accuracy	Epistemic (MI) Accuracy
Number Theory	62	71.0% (44/62)	74.2% (46/62)
Prealgebra	82	53.7% (44/82)	47.6% (39/82)
Precalculus	56	16.1% (9/56)	19.6% (11/56)

The difficulty level breakdown (Table 3) also shows a similarly mixed pattern. The MI model has improved on both easier and medium-difficulty problems, increasing from 72.1% to 74.4% on Level 1, from 62.2% to 65.6% on Level 2, and from 40.6% to 43.0% on Level 4. The two models were tied at 53.3% on Level 3. However, at the hardest level, the MI model underperformed the baseline, decreasing from 26.9% to 22.4% on Level 5. This suggests that epistemic selection may help most when the model can still benefit from better examples, but it may have less influence on the hardest problems where both models remain far from reliable.

Table 3. MATH-500 accuracy by difficulty level

Difficulty Level	Total	Baseline Accuracy	Epistemic (MI) Accuracy
Level 1	43	72.1% (31/43)	74.4% (32/43)
Level 2	90	62.2% (56/90)	65.6% (59/90)
Level 3	105	53.3% (56/105)	53.3% (56/105)
Level 4	128	40.6% (52/128)	43.0% (55/128)
Level 5	134	26.9% (36/134)	22.4% (30/134)

The most significant interpretation is that the two selection rules produced very different training datasets while resulting in only small downstream differences. Across the 12,000 training problems, the total-entropy and MI rules selected different trajectories for 8,105 problems, or 67.54% of the dataset. In other words, the uncertainty metric change, which examples were originally chosen for fine-tuning on the baseline model. Despite this large dataset-level difference, the final benchmark gap was very small. This concludes that many alternative teacher trajectories may be similarly useful for supervised fine-tuning, or that the current evaluation may be too limited to show meaningful differences in reasoning quality.

Conclusion

In this study, the results provide modest evidence for using mutual information as a data trajectory selection method. The MI-based model performed slightly better than the total-entropy model on both MATH-500 and GSM8K, which suggests that epistemic uncertainty may help select reasoning data. However, the gains were very small, so the results are not

strong enough to claim that MI is clearly better than total predictive entropy.

In conclusion, Bayesian epistemic uncertainty is a promising technique for reasoning trajectory selection, but in this project, it produced only modest improvements. The results suggest that MI-based selection is worth further research with stronger evaluation methods, multiple training runs, and broader experiments.

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